An Empirical Study on Political Polarization within the YouTube Ecosystem : Focusing on Linguistic Divergence and Network Isolation.

Yuri Kim

09 April 2026

Table of Contents

1. Introduction **3**

2. Research Question **3**

3. Data Collection **4**

4. Data Preprocessing **4**

5. Methodology **5**

- **5.1 Linguistic Analysis** 5
- **5.2 Network Analysis** 5
- **5.3 Integrating Linguistic and Network Perspectives** 5

6. Results **5**

- **6.1 Title-Based Analysis (Trump-filtered)** 5
- **6.2 Title-and-Comment Analysis (Trump-filtered)** 6
- **6.3 Analysis without Keyword Filtering** 6
- **6.4 Network Analysis** 6
- **6.5 Synthesis** 7

7. Data Collection **7**

8. Conclusion **8**

9. References **8**

10. LLM Declaration **8**

1. Introduction

Online platforms have become central spaces for political communication, where information is not only distributed but also actively interpreted and reshaped by users. In particular, YouTube provides a unique environment in which political discourse emerges through both content production and audience engagement, as videos and user comments jointly reflect how political issues are framed, discussed, and emotionally expressed. Language plays a critical role in political communication, as patterns of word choice often reflect underlying ideological positions (Monroe et al., 2017). Distinct vocabularies can signal differences in political identity and framing, allowing computational methods to detect semantic variation across groups (Demszky et al., NAACL 2019). At the same time, emotional expression is deeply embedded in political discourse, with sentiment and affective language shaping how issues are perceived and evaluated (Brader, T., 2006). Beyond linguistic patterns, the structure of relationships between content producers can reveal broader dynamics of political alignment (Barbera, P., 2015). Network-based analyses have consistently found that political communication tends to form ideologically clustered communities, often characterized by homophily and the emergence of echo chambers (Bakshy et al., 2015). Such structural patterns provide a complementary perspective to text-based analysis, capturing how actors are positioned within the broader communication landscape. Building on these perspectives, this study examines whether ideologically distinct YouTube channels exhibit systematic differences in language use and whether such differences correspond to structural clustering in the channel network. By jointly analyzing textual features and inter-channel relationships, this study aims to provide a more comprehensive understanding of how political polarization manifests in online video-based discourse.

2. Research Question

This study aims to investigate the extent to which political content on YouTube exhibits polarization across both linguistic and structural dimensions. As online platforms increasingly shape how political information is produced and consumed, it is important to examine not only differences in opinions, but also how these differences are reflected in language use and interaction patterns among ideological groups.

To this end, the analysis is guided by the following research questions:

RQ1. *Do ideologically distinct YouTube channels exhibit systematic differences in their language use when discussing a shared political issue?*
This question focuses on identifying semantic and emotional variation across channels, examining whether differences in vocabulary, phrasing, and sentiment reflect underlying ideological positions.

RQ2. *Do these linguistic differences correspond to structural clustering in the channel network?*
This question investigates whether channels that are linguistically similar are also structurally connected, forming ideologically cohesive communities within the network.

Together, these questions aim to assess whether differences in discourse are not only observable at the level of language but also reinforced through network-level organization, providing a more comprehensive understanding of political polarization in online video platforms.

3. Data Collection

This study constructs a dataset of political content on YouTube using the YouTube Data API v3. The data collection process was guided by two principles: ensuring ideological diversity and enabling controlled comparison through a shared political topic. To identify relevant channels, a multi-query search strategy was used, combining results from multiple political search terms (e.g., "political commentary US," "right wing politics," "liberal news"). The top 50 channels per query were retrieved and deduplicated, yielding 398 unique channels. To focus on influential content, only channels with more than one million subscribers were retained, resulting in 27 channels. Channels were manually labeled as progressive or conservative based on their content and positioning. To ensure clear ideological distinction, channels with ambiguous orientations were excluded, leaving a final set of six channels (three per group). For each channel, the 50 most recent videos were collected. To control for topic variation, only videos containing keywords such as "Trump," "Donald Trump," or "President" were retained. Initially, video descriptions were included, but were removed due to repetitive and non-informative content. Instead, user-generated comments were incorporated to capture audience-level discourse. For each video, up to 30 top-level comments were collected based on relevance ranking. The final dataset consists of video titles and associated comments from ideologically distinct channels within a shared topic, enabling analysis of linguistic patterns while minimizing topic-related confounding effects.

4. Data Preprocessing

The collected textual data were preprocessed to ensure consistency and suitability for downstream linguistic analysis. All text data, including video titles and user comments, were first converted to lowercase and cleaned by removing punctuation, numerical values, and non-alphabetic characters. The cleaned text was then tokenized into word-level units. To focus on meaningful linguistic patterns, common stopwords were removed using both standard stopword lists and a custom-defined set tailored to the dataset. The custom stopwords included platform-specific and non-informative terms (e.g., "subscribe," "channel," "video") as well as frequently occurring proper names that do not contribute to distinguishing ideological content. To further capture contextual relationships between words, bigram features were constructed by combining adjacent tokens. During this process, bigrams containing stopwords or non-informative terms were filtered out to retain only meaningful word combinations. This allows the analysis to identify recurring phrases and framing patterns that may not be observable at the unigram level. In addition, infrequent tokens were removed by applying a minimum frequency threshold across the dataset, reducing noise and improving the stability of subsequent statistical measures such as TF-IDF. Following preprocessing, the text was aggregated at the channel level, where all titles and associated comments for each channel were combined into a single document. This

representation enables the analysis to capture the overall linguistic and framing characteristics of each channel, rather than focusing on individual videos in isolation.

5. Methodology

This study employs a combined linguistic and network-based framework to examine political polarization in YouTube discourse. By integrating textual analysis with network modeling, the methodology captures both how content is expressed and how channels are structurally related.a

5.1 Linguistic Analysis

To examine differences in language use across ideological groups, this study applies TF-IDF, bigram analysis, and sentiment analysis. TF-IDF is used to identify distinctive terms for each group by emphasizing relatively unique words while down-weighting common ones, enabling detection of differences in vocabulary and framing. Bigram analysis captures recurring word pairings, allowing the identification of phrase-level patterns that better reflect how meaning is constructed in political discourse. Sentiment analysis, based on a lexicon approach, measures differences in emotional tone by comparing positive and negative word usage across groups. Together, these methods capture linguistic variation at the levels of vocabulary, framing, and emotion.

5.2 Network Analysis

To complement the linguistic analysis, a similarity-based network is constructed in which channels are represented as nodes and edges are defined by cosine similarity of their textual content. This network reflects linguistic proximity between channels. Community structure is examined using the Louvain algorithm, which identifies clusters by maximizing modularity. In addition, ideological assortativity is computed to assess whether channels are more likely to connect with others sharing the same ideological label.

5.3 Integrating Linguistic and Network Perspectives

Finally, linguistic patterns are linked to network structure by examining whether channels with similar language use are also structurally grouped. This integrated approach enables the study to assess polarization at both the level of discourse and network organization.

6. Results

6.1 Title-Based Analysis (Trump-filtered)

As a baseline, analysis using titles alone shows limited but observable differences. Progressive channels emphasize conflict-related terms (e.g., *"erupts," "brutal," "collapse"*), while

conservative channels focus more on events and figures (e.g., *"celebration," "cabinet," "Melania"*). Sentiment shows a clear contrast: progressive titles are strongly negative (47 negative vs. 4 positive), whereas conservative titles are more positive (10 positive vs. 6 negative). However, titles provide only a surface-level view due to their brevity.

6.2 Title-and-Comment Analysis (Trump-filtered)

With comments included, differences become clearer. Unigram results show distinct vocabularies, with conservative discourse including terms such as *"bless," "reuters,"* and *"smartmatic,"* while progressive discourse includes *"insane," "muslims,"* and *"criminal."* Bigram analysis reveals stronger framing differences. Conservative channels use phrases such as *"god bless," "propaganda trick,"* and *"oil prices,"* combining evaluative language with issue references. Progressive channels use more critical expressions such as *"destroyed biden's"* and *"war crimes."* Sentiment further reinforces this divide: progressive discourse is strongly negative (1086 negative vs. 507 positive), while conservative discourse is more balanced (366 negative vs. 395 positive). Overall, comments reveal clearer differences in framing and tone than titles alone.

6.3 Analysis without Keyword Filtering

Without topic restriction, differences persist but become more general. Progressive discourse emphasizes international conflict and normative judgment (e.g., *"war crimes," "human rights"*), while conservative discourse focuses on domestic policy and identity (e.g., *"save america," "voter id"*). Sentiment remains more negative for progressive channels (1793 negative vs. 832 positive) compared to conservatives (670 negative vs. 515 positive). These results suggest that polarization is not limited to a single topic, though it is amplified in Trump-related content.

6.4 Network Analysis

The network shows no clear community structure (modularity = 2.076521e-16). However, channels within the same ideology are slightly more similar (0.170) than cross-ideological pairs (0.124), indicating weak but present ideological alignment. As shown in Figure 1, channels are not separated into distinct clusters, and connections are distributed across groups. This suggests that while ideological similarity exists, it does not result in strong structural separation.

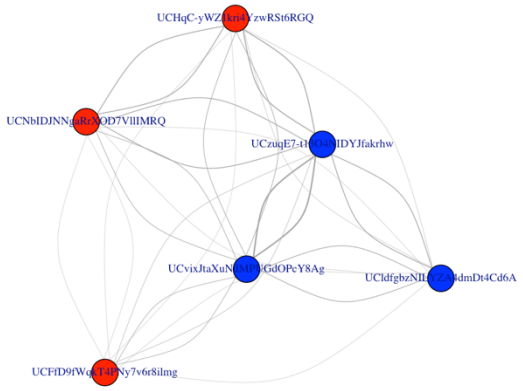

Figure 1. Similarity-based network of YouTube channels. Nodes represent channels colored by ideology (red = progressive, blue = conservative), and edges indicate textual similarity.

6.5 Synthesis

Overall, polarization is more evident in language than in network structure. Ideological groups differ clearly in vocabulary, framing, and sentiment, but these differences do not translate into strongly separated communities. This suggests that polarization operates primarily through discourse within a shared communication space rather than through fully isolated echo chambers.

7. Data Collection

This study has several limitations.

First, the classification of channels into ideological categories relies on manual labeling. Although this approach allows for careful interpretation of channel content and positioning, it introduces a degree of subjectivity. In practice, political content is not always strictly binary, and some channels may exhibit mixed or shifting ideological orientations that are not fully captured by a left–right classification.

Second, the dataset exhibits imbalance across ideological groups, particularly in the Trump-filtered subset of videos. Differences in the number of videos and associated comments between progressive and conservative channels may influence the observed patterns in both linguistic and sentiment analyses. As a result, some findings may reflect underlying data distribution differences rather than purely ideological effects.

Third, the network analysis is based on textual similarity rather than direct interaction between channels. As such, the constructed network does not capture actual relationships such as audience overlap, recommendation pathways, or subscription behavior. This limits the ability to

draw strong conclusions about the presence or absence of echo chambers in a structural sense, as the network reflects similarity in discourse rather than real patterns of information flow.

8. Conclusion

This study examined political polarization in YouTube discourse by integrating linguistic and network-based analyses. The results show that ideological differences are clearly reflected in language use, including vocabulary, framing, and sentiment, with progressive and conservative channels exhibiting distinct patterns of expression. At the same time, the network analysis reveals only weak structural separation, suggesting that these differences do not fully translate into clearly segregated communities. Instead, the findings indicate that polarization is more strongly manifested in how political issues are interpreted and expressed than in how channels are structurally organized. Overall, this study highlights that polarization in online platforms may operate as a divergence in discourse within a shared communication space, rather than as complete fragmentation into isolated echo chambers.

9. References

Dorottya Demszky, Nikhil Garg, Rob Voigt, James Zou, Jesse Shapiro, Matthew Gentzkow, and Dan Jurafsky. 2019. Analyzing Polarization in Social Media: Method and Application to Tweets on 21 Mass Shootings. https://doi.org/10.48550/arXiv.1904.01596

Brader, Ted. Campaigning for Hearts and Minds: How Emotional Appeals in Political Ads Work. University of Chicago Press, 2006.

Barberá, Pablo. "Birds of the Same Feather Tweet Together: Bayesian Ideal Point Estimation Using Twitter Data." Political Analysis, vol. 23, no. 1, 2015, pp. 76–91. JSTOR, http://www.jstor.org/stable/24572842.

Eytan Bakshy *et al.*, Exposure to ideologically diverse news and opinion on Facebook. Science348,1130-1132, 2015.DOI:10.1126/science.aaa1160

Monroe BL, Colaresi MP, Quinn KM. Fightin' Words: Lexical Feature Selection and Evaluation for Identifying the Content of Political Conflict. *Political Analysis*. 2017;16(4):372-403. doi:10.1093/pan/mpn018

10. LLM Declaration

Large language model tools were used in this project in a limited and supportive capacity. ChatGPT was used for language support, including proofreading, improving clarity, and assisting with debugging. GitHub Copilot was used for code suggestions, auto-completion, and comment generation during implementation. All research design, data collection, data analysis, interpretation of results, and conclusions were conducted by me. I take full responsibility for the content of this work.